

Bayes Classifier

Bayes Classifier

$$P(y|x) = \frac{P(y)P(x|y)}{P(x)}$$

$$y_{MAP} = \arg \max_{y \in Y} \frac{P(y)P(x|y)}{P(x)} = \arg \max_{y \in Y} P(y)P(x|y)$$

$$\arg \max_{y \in Y} P(y)P(x|y) = \arg \max_{y \in Y} P(y)P(x_1, x_2, \dots, x_n | y)$$

Naïve assumption (features are independent):

$$P(x_1, x_2, \dots, x_n) = P(x_1|y) \cdot P(x_2|y) \cdot \dots \cdot P(x_n|y)$$

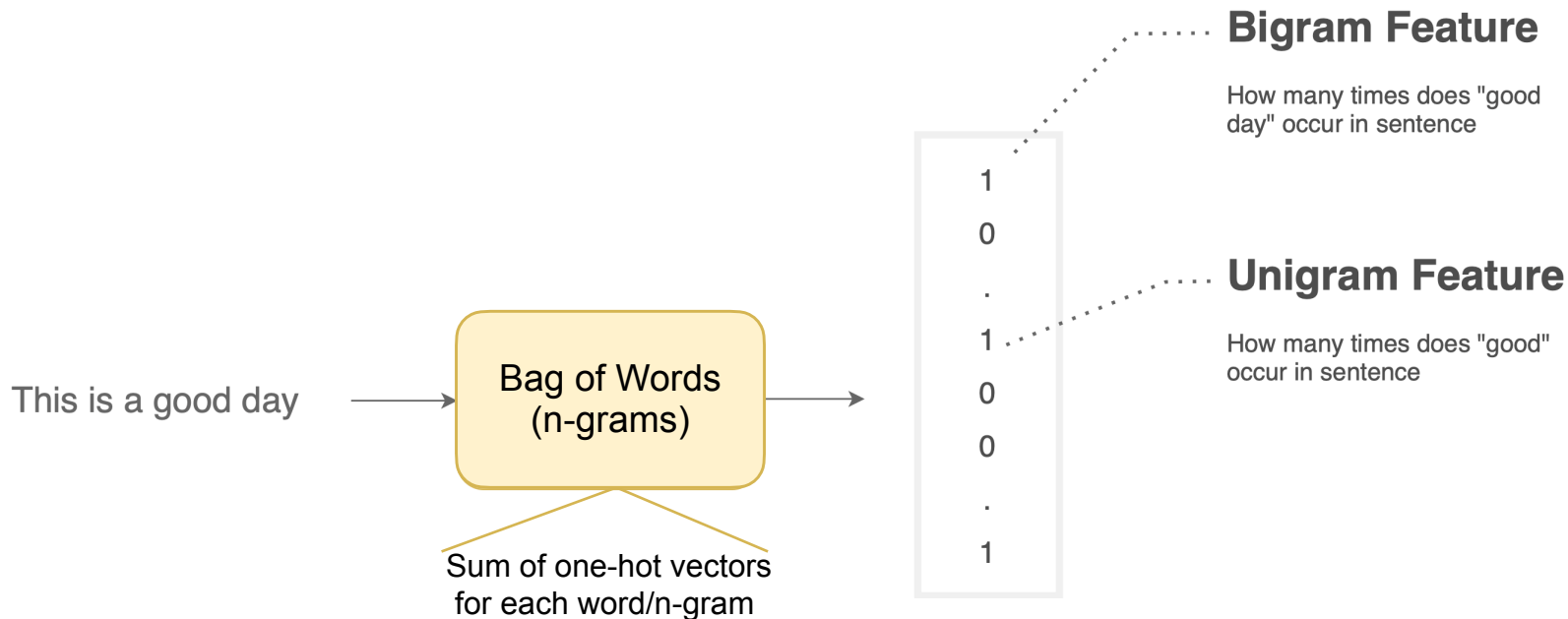
Naïve Bayes Classifier

$$P(x_1, x_2, \dots, x_n) = P(x_1 | y) \cdot P(x_2 | y) \cdot \dots \cdot P(x_n | y)$$

$$y_{MAP} = \arg \max_{y \in Y} P(y)P(x | y)$$

$$y_{NB} = \arg \max_{y \in Y} P(y) \prod_i P(x_i | y)$$

NB for Text: Bag of Words



Words in documents

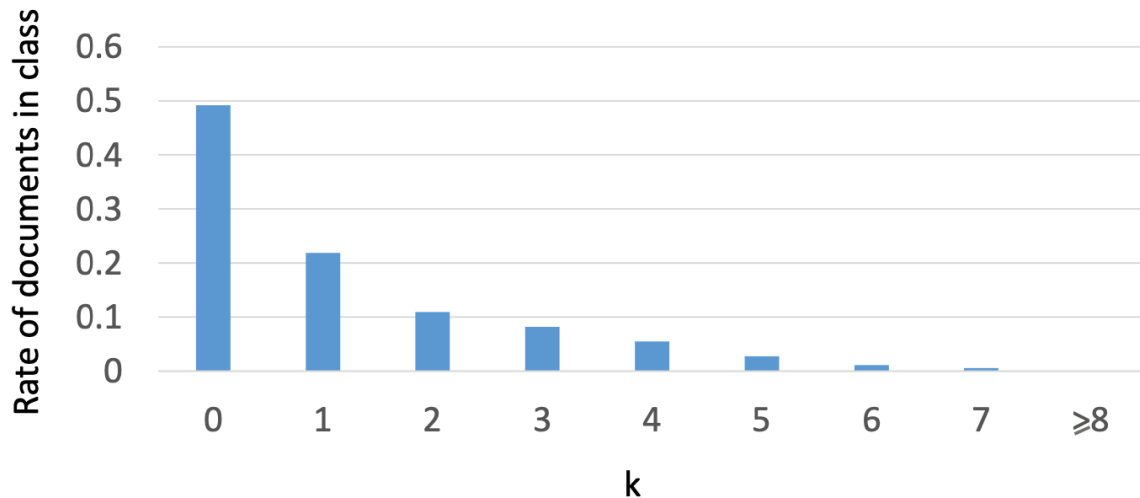
$$y_{NB} = \arg \max_{y \in Y} P(y) \prod_i P(x_i | y)$$

- $P(y)$ – frequency of class y
- $P(x_i = k | y)$ – probability of value k of feature x_i in class y . Such as a proportion of documents in class with $x_i = k$.

$$P(x_i | y_j) = \frac{\text{count}(x_i, y_j)}{\text{count}(y_j)}$$

Problem – $\text{count}(x_i, y_j) = 0$

$$\hat{P}(x_i | y_j) = \frac{\text{count}(x_i, y_j) + 1}{\text{count}(y_j) + K}$$



Bayes classifier with various types of features

Binary features:

$$P(x_i | y) = P(x_i = 1 | y)x_i + (1 - P(x_i = 1 | y))(1 - x_i), \quad x_i \in \{0, 1\}$$

Distribution for continuous features:

$$p(x_i | y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} e^{-\frac{(x_i - \mu_y)^2}{2\sigma_y^2}}$$

Distribution evaluation

A simple option is to use sample mean and variance for μ and σ .



A more complex option is to use expectation-maximization algorithm (EM) with Gaussian mixtures

Expectation-Maximization (EM)

Define Gaussian mixture of K distributions by:

μ_k – mean vector, Σ_k – covariance matrix

α_k – a weight of distribution k , a probability that a random point belongs to it

$$\sum \alpha_k = 1$$

Affinity of object x_i to distribution k :

$$w_{ik} = \frac{p(x_i | \mu_k, \Sigma_k) \cdot \alpha_k}{\sum_j p_j(x_i | \mu_j, \Sigma_j) \cdot \alpha_j}$$

E-step: calculate w_{ik}

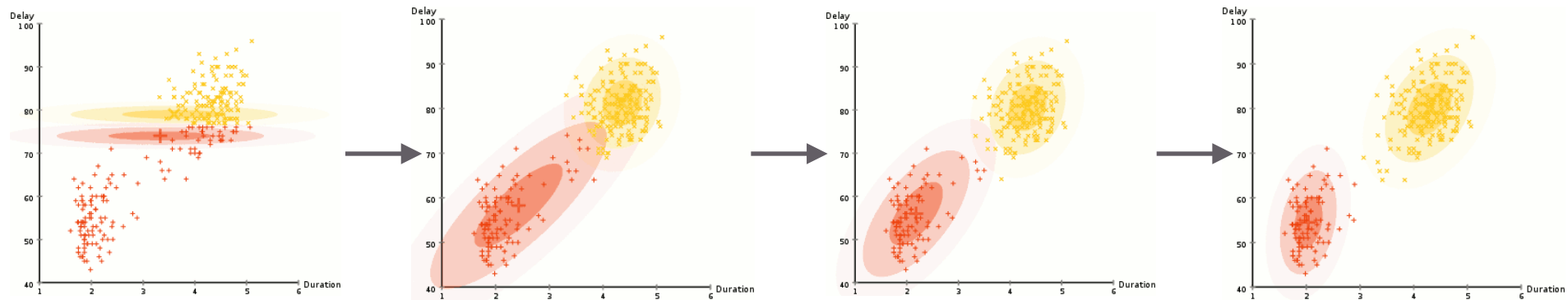
M-step:

$$\alpha_k^{new} = \frac{\sum_{i=1}^N w_{ik}}{N} = \frac{N_k}{N}$$

$$\mu_k^{new} = \frac{1}{N_k} \sum_{i=1}^N w_{ik} \cdot x_i$$

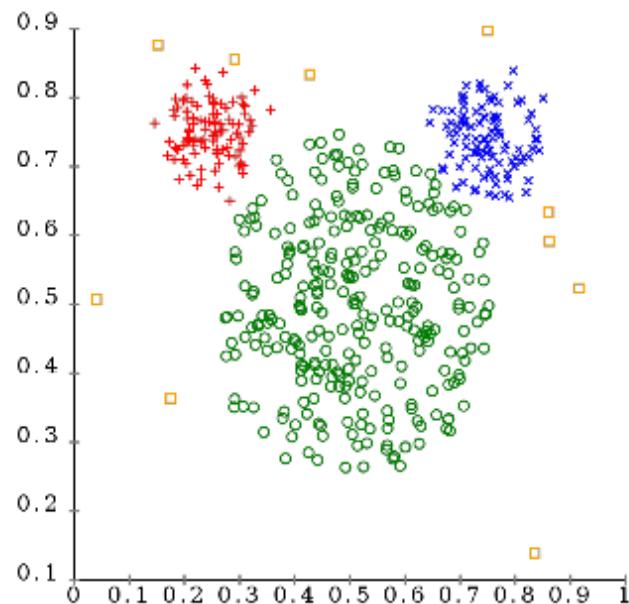
$$\Sigma_k^{new} = \frac{1}{N_k} \sum_{i=1}^N w_{ik} (x_i - \mu_k^{new})(x_i - \mu_k^{new})^T$$

EM Clustering

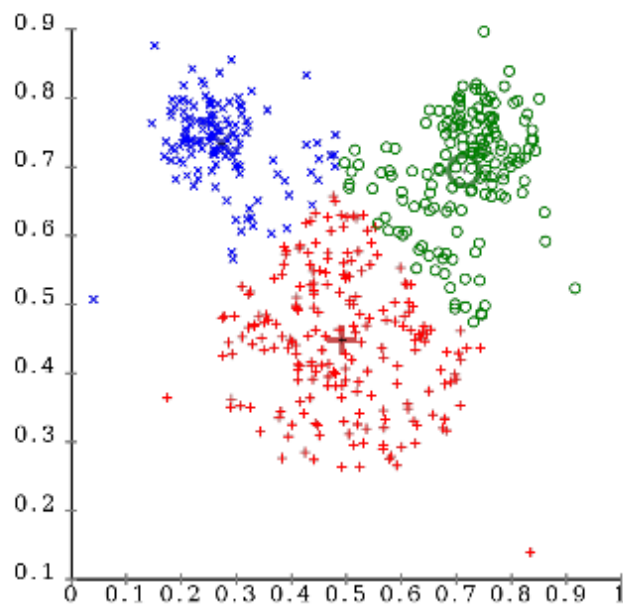


EM Clustering

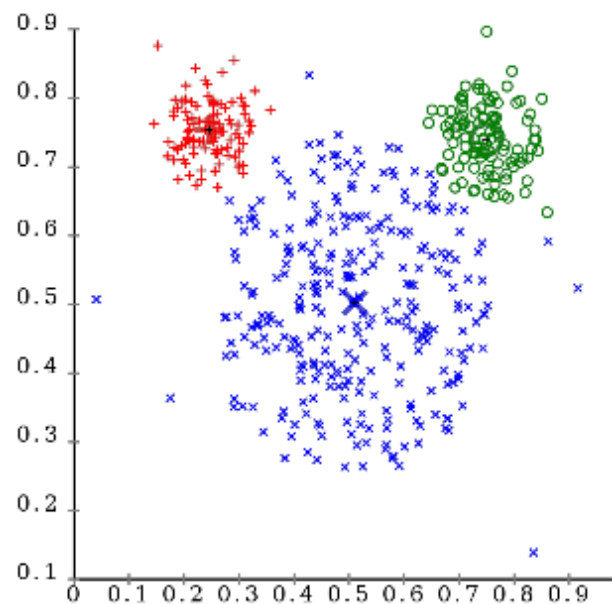
Original Data



k-Means Clustering



EM Clustering



Naïve Bayes is a very nice baseline!